



Predicting Total Power Output in Wave Energy Farms

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Introduction

Wave energy converters (WECs) generate electricity from ocean wave motion. This study evaluates six supervised regression models: Linear Regression, Ridge, Lasso, Random Forest, Extra Trees, and XGBoost to determine which approach most accurately predicts total farm power output from individual WEC outputs. The dataset used is Large-scale Wave Energy Farm WEC_Perth_49, containing 36,043 observations across 149 features derived from a wave energy farm in Perth, Australia.

Methodology

1. Exploratory Data Analysis (EDA):

- Correlation analysis showed that Power1-Power49 were strongly associated with Total Power (Fig.1).
- XY coordinates showed weak correlations with Total Power. As a result, they were removed during feature selection.

2. Feature Selection:

- Only Power1-Power49 were retained based on strong correlations with Total Power.
- XY coordinates were excluded due to weak relationships.
- Final Feature set reflects the linear characteristics of the dataset. WEC power generation directly influences the total power output of the Wave Energy Farm.

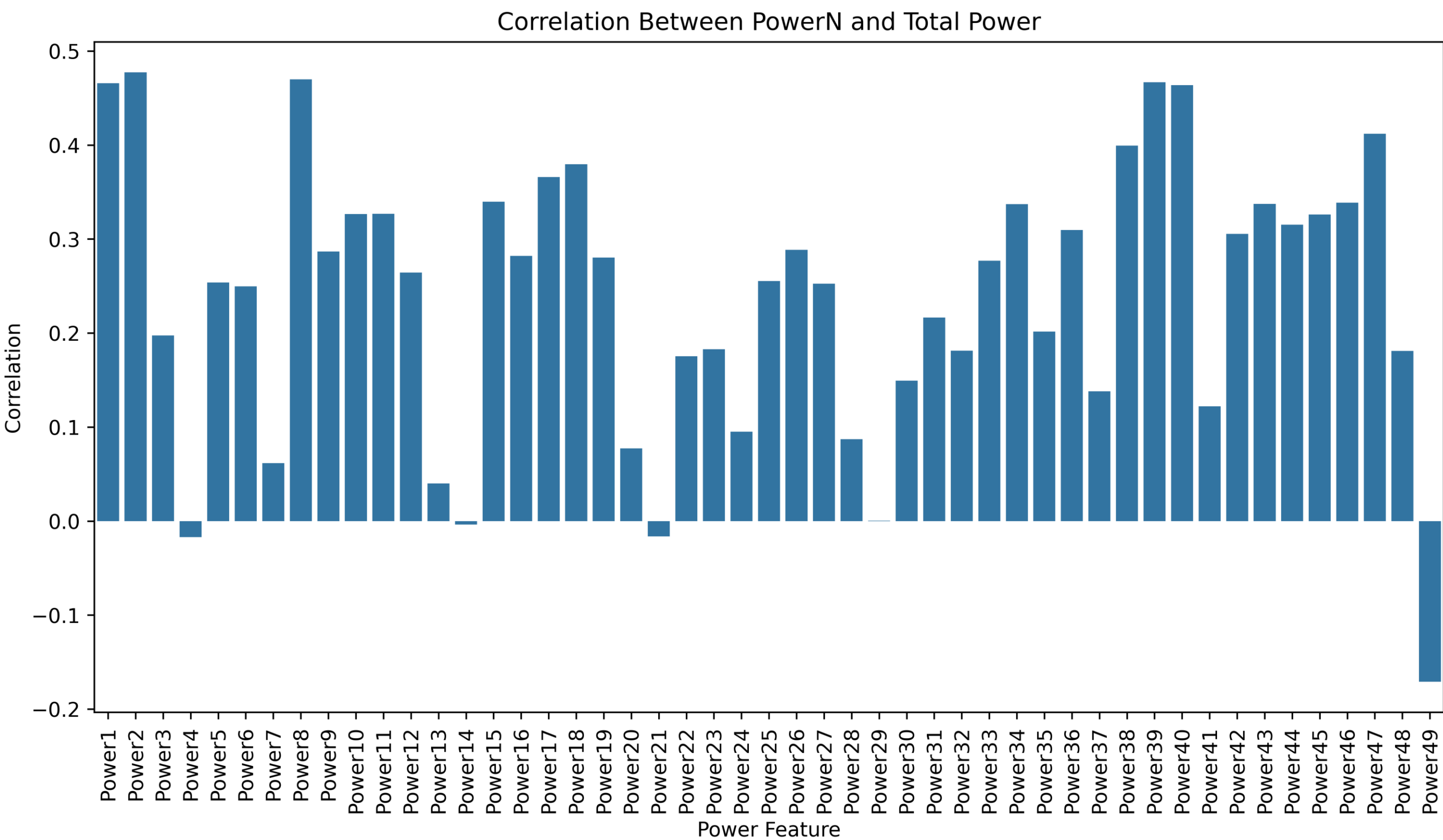


Figure 1: Correlation between each Power(n) variable and the Total Power target

3. Preprocessing:

- All Power(n) features were standardized using StandardScaler
- Scaling ensured fair comparison across linear and tree-based models.
- No dimensionality reduction was applied to help interpretability.

4. Train/Test Split:

- The dataset was split into 80/20 train/test split which was used by all models for consistency.

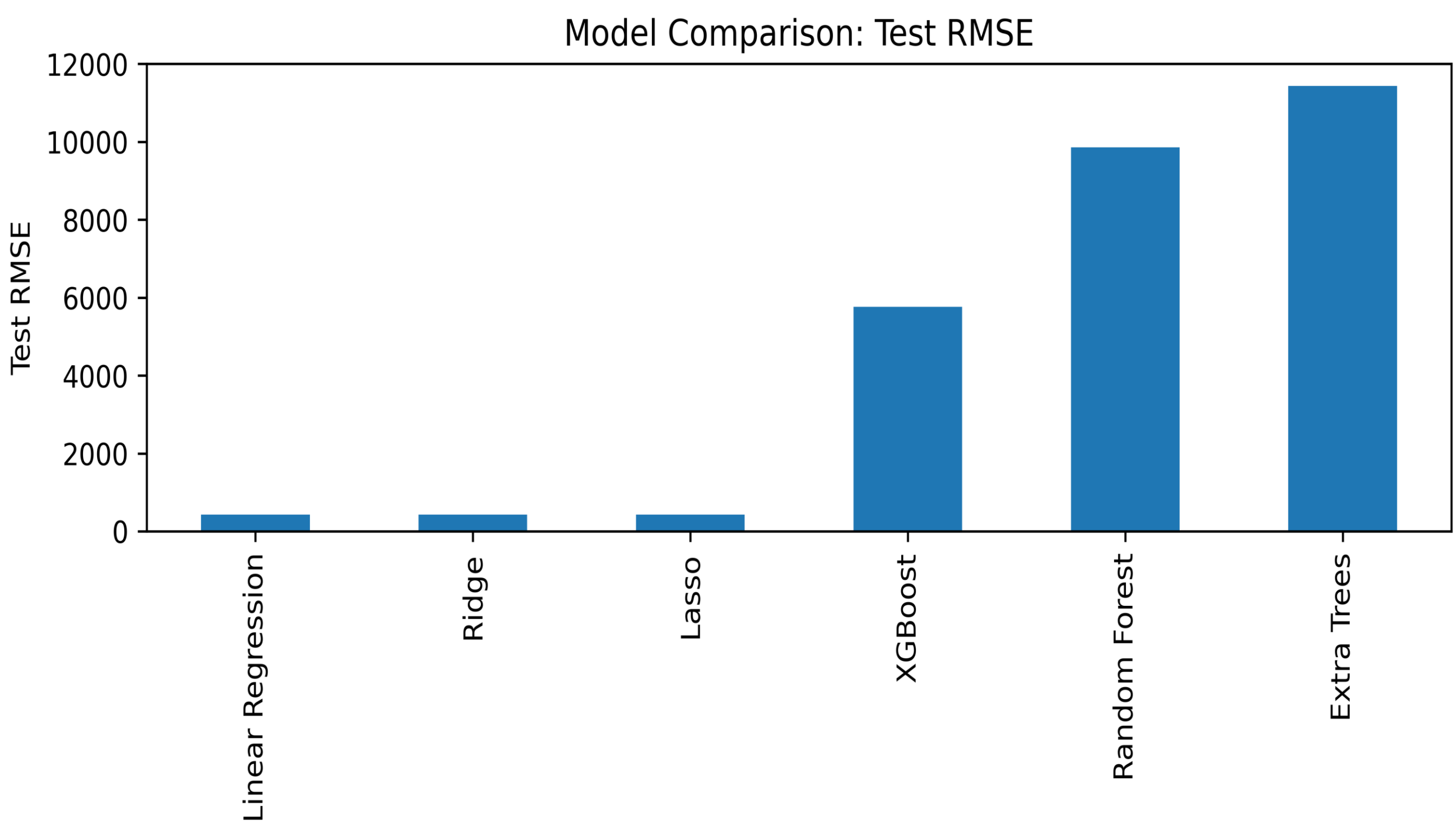


Figure 2: Plotting Test RMSE for each model to show average prediction error by model, measured in watts.

5. Hyperparameter Tuning:

- GridSearchCV was used for all models.
- Linear models used a 5-fold CV
- Tree-based models used a 3-fold CV due to higher computational cost.
- Scoring metric: Negative Mean Squared Error (NMSE)

6. Evaluation Metric:

- Final performance reported using Test RMSE (Fig.2).
- RMSE was chosen for interpretability in physical units (watts).
- Target magnitude is measured in hundreds of thousands of watts, so relative error is small.

Results

- Linear Regression ranked #1 in Table 1 with ~426 W test RMSE
- Figure # shows the performance gap favoring linear models.
- Tree-based models overfitted and generalized poorly.

Rank	Model	Test MAE	Test MSE	Test RMSE	Test R2
1	Linear Regression	289.25	181428.84	425.94	0.999988
2	Ridge	289.13	181453.4	425.97	0.999988
3	Lasso	289.2	181507.33	426.04	0.999988
5	Random Forest	3318.28	97264147.46	9862.26	0.993401
6	Extra Trees	4263.33	130759262.6	11435	0.991128
4	XGBoost	2593.55	33234961.98	5764.98	0.997745

Figure 3: Ranking of all models in comparison, scored by Test RMSE

Conclusions

- The relationship between Power(n) features and Total Power is strongly linear. As a result, linear models performed better than tree-based models across all metrics.
- Linear Regression delivered the best generalization performance across all models.
- Tree-based models overfitted despite extensive hyperparameter tuning.

Future Work

- Apply log-scaling or other target transformations to address heteroscedasticity at upper Power values.
- Test models on the remaining farm configurations for Perth and Sydney to assess whether the models still perform identically given different locations and data.
- Explore using the XY coordinate data as an input feature.